

Polynomial Regression for Geothermal Systems

Machine Learning Assignment 1

Roll number: BT2024128

Problems: Steam turbine optimization (var1), thermal reservoir mapping (var2)

Primary metrics: Mean Squared Error (MSE), coefficient of determination (R^2)

GitHub repository: https://github.com/oki-dokii/ML-Assignment1_BT2024128

Abstract

Two polynomial regression models were developed for the personalized datasets assigned to BT2024128. Feature subsets were screened with ordinary least squares (OLS), then regularized models were compared using cross-validation. The selected var1 model expands all six inputs to degree 5, uses Lasso to select terms, and refits them with Ridge; var2 expands all three inputs to degree 12 and uses Ridge alone. Repeated five-fold cross-validation produced MSE 0.2975 and $R^2 = 0.9724$ for var1, and MSE 0.2291 and $R^2 = 0.9945$ for var2.

1 Dataset and task

Each problem contains 1,000 training observations and 1,000 test observations. All columns are numeric, all input variables lie between -1 and 1 , and no values are missing. The training sets contain the continuous target y ; the test targets are hidden.

Problem	Training shape	Test shape	Candidate inputs
var1	1000×7	1000×6	$x_1, \dots, x_6$
var2	1000×4	1000×3	x_1, x_2, x_3

The train and test files for each problem were kept separate. Test values were used only after the final configuration had been selected and fitted on the complete corresponding training set.

2 Polynomial regression pipeline

For p selected inputs and maximum degree d , the expansion contains every monomial whose total degree is at most d . Excluding the constant term, the number of generated terms is

$$N(p, d) = \binom{p+d}{d} - 1.$$

The prediction model is linear in these expanded terms:

$$\hat{y} = \beta_0 + \sum_{j=1}^{N(p,d)} \beta_j \phi_j(\mathbf{x}).$$

Both scikit-learn pipelines apply `PolynomialFeatures`, then `StandardScaler`. Var1 next uses Lasso ($\alpha_L = 0.03$) to retain terms with nonzero coefficients, then Ridge ($\alpha_R = 0.1$) to refit those terms. Lasso minimizes squared error plus an L_1 coefficient penalty, encouraging sparse selection. Var2 uses Ridge alone ($\alpha_R = 1$). The Ridge stage minimizes

$$\sum_i (y_i - \beta_0 - \mathbf{z}_i^\top \boldsymbol{\beta})^2 + \alpha_R \|\boldsymbol{\beta}\|_2^2,$$

where $\mathbf{z}_i$ contains the selected standardized polynomial terms. Ridge shrinks large coefficients and helps control variance. The intercept is not penalized. Expansion, scaling, and var1 term selection are all fitted inside each validation fold to prevent leakage.

3 Model-selection procedure

The following procedure was applied independently to var1 and var2.

1. Screen every non-empty feature subset using OLS polynomial regression. Degrees up to 10 for var1 and 20 for var2 were considered, omitting OLS designs with more than 700 terms against 800 fitting observations per fold.
2. Keep the full feature sets because they had the lowest OLS validation error by a large margin.
3. Search Ridge models on these feature sets: degrees 3–6 and $\alpha_R \in \{0.1, 1, 3, 10, 30, 100\}$ for var1; degrees 7–16 and $\alpha_R \in \{0.01, 0.1, 1, 3, 10\}$ for var2.
4. For var1 degree 5, also search Lasso term-selection strengths $\alpha_L \in \{0.003, 0.01, 0.03\}$ followed by Ridge strengths $\alpha_R \in \{0.1, 1, 10, 30\}$. Keep var2 Ridge-only because tested Lasso selection did not lower its five-fold MSE.
5. Compare MSE (primary) and R^2 (secondary) using shuffled five-fold cross-validation. Check the var1 tuning procedure with a separate outer five-fold, inner three-fold nested validation.
6. Re-evaluate each fixed final configuration using five repeats of five-fold cross-validation.

Ridge permits more polynomial terms than OLS because its penalty stabilizes coefficient estimation. Lasso removes weak or redundant degree-5 terms in var1. All shuffled folds use fixed seeds, and the test targets were never used for these decisions.

3.1 Degree comparison

Table 1: Five-fold cross-validation of representative configurations

Problem	Model	Degree	α_L	α_R	Mean MSE
var1	OLS	4	–	0	0.7725
var1	OLS	5	–	0	1.0671
var1	Ridge	5	–	10	0.4265
var1	Lasso + Ridge	5	0.01	0.1	0.3235
var1	Lasso + Ridge	5	0.03	0.1	0.3082
var2	OLS	8	–	0	0.2544
var2	Ridge	10	–	1	0.2346
var2	Ridge	12	–	1	0.2268
var2	Ridge	13	–	1	0.2307

For var1, Lasso selection plus Ridge improves on Ridge alone at degree 5. The nested procedure averaged MSE 0.3086 and $R^2 = 0.9714$ across its outer folds; four of five inner searches chose $\alpha_L = 0.03$, $\alpha_R = 0.1$, while one chose $\alpha_R = 1$. For var2, degree-12 Ridge remained best among the tested candidates.

4 Feature selection

The preliminary OLS screen favored all available inputs for both datasets. The best five-feature var1 alternative (x_1, x_2, x_3, x_5, x_6 , degree 3) had MSE 1.8294, compared with 0.7725 for the full-feature degree-4 OLS baseline. All lower-dimensional var2 subsets were far behind the full three-feature degree-8 OLS model. Thus input variables were retained, while var1’s Lasso stage selected among their expanded polynomial terms. It retained 69 of 461 terms after the final full-data fit.

5 Final validation results

Table 2: Repeated five-fold validation of the selected models (five repeats)

Problem	Final model	Degree	Retained terms	MSE	R^2
var1	Lasso + Ridge	5	69 of 461	0.2975 ± 0.0296	0.9724 ± 0.0036
var2	Ridge	12	454	0.2291 ± 0.0251	0.9945 ± 0.0013

The repeated estimates are close to the original five-fold screening scores, which shows that the choices are not tied to one random split. A separate five-fold out-of-fold prediction run gave the following results.

Problem	OOF MSE	OOF R^2	Training MSE	Training R^2
var1	0.3082	0.9717	0.2360	0.9783
var2	0.2268	0.9947	0.1487	0.9965

5.1 Diagnostic interpretation

For var1, out-of-fold predictions track the identity line across negative and positive scores. Residuals are centered near zero, with a few larger deviations at extreme responses. Selecting a sparse set of degree-5 terms reduced five-fold MSE from 0.4265 for Ridge-only to 0.3082. The 69-term count above comes from fitting all 1,000 labeled rows; each validation fold selected terms independently.

For var2, out-of-fold predictions lie close to the identity line over the observed target range, approximately -30 to 39 . Residuals remain centered around zero without a strong systematic curve. Degree 12 captures more structure than the earlier degree-8 OLS baseline while Ridge limits unstable coefficients.

6 Final training and test inference

After selection, each pipeline was refitted on all 1,000 observations from its training dataset. Predictions were generated in the original test-row order. The final output ranges were -10.1324 to 16.6722 for var1 and -29.7657 to 38.7586 for var2. Both submission files follow the provided sample: exactly 1,000 finite predictions in a single column named `y`, with no exported row index.

7 Validation plots

The bars compare five-fold validation MSE; shorter bars are better. They show why the final model for each dataset was chosen. The full degree-search and residual-diagnostic plots are also saved in the GitHub repository.

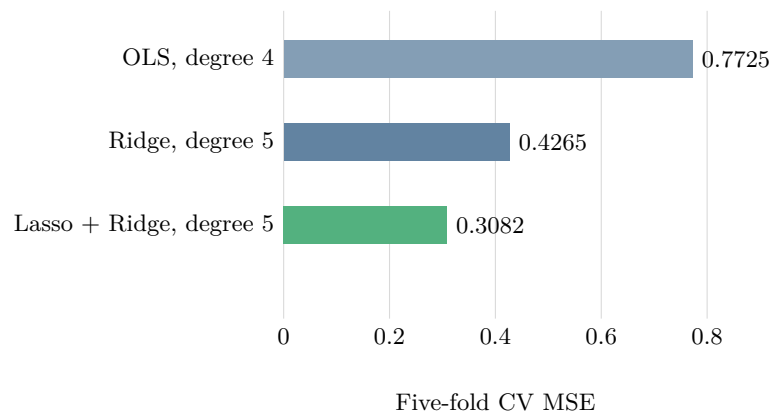


Figure 1. var1: Lasso term selection followed by Ridge reduces validation error.

For var2, the selected degree-12 Ridge model also improves on the best OLS baseline. Lasso selection was tested but did not further reduce error on this dataset.

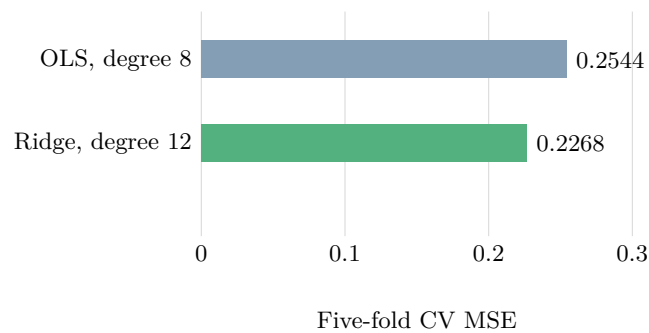


Figure 2. var2: degree-12 Ridge improves on ordinary least squares.

8 Reproducibility and safeguards

The implementation provides separate scripts for selection and inference. The selection script writes OLS, Ridge, and Lasso-plus-Ridge search tables, nested and repeated-validation metrics, a degree-selection figure, and out-of-fold diagnostic plots. The prediction script validates schemas, rejects missing or nonnumeric values, checks prediction length and finiteness, saves the fitted pipelines, and writes the two submission CSV files.

- **No test leakage:** Neither test inputs nor hidden targets influence feature or degree selection.
- **Pipeline isolation:** Polynomial generation, scaling, and Lasso selection are refitted inside every validation fold.
- **Reproducible splits:** All shuffled validation procedures use random seed 42.
- **Complexity control:** OLS screening avoids expansions larger than 700 terms; Lasso sparsifies var1's terms and Ridge shrinks its retained coefficients.
- **Submission checks:** Output row counts, column name, order, and finite numeric values are verified programmatically.

9 Limitations

The true test targets are hidden, so expected performance is estimated from cross-validation rather than directly measured. Polynomial regression can also become sensitive near the edges of the input domain, particularly for the six-variable var1 model. The supplied train and test inputs share the same bounded range of $[-1, 1]$, which limits extrapolation risk, but some test points combine boundary values more frequently than the training data. Predictions at those combinations may extend beyond the observed training-target range.

Cross-validation is an estimate, not a measurement against the hidden test targets. Because the same training set informed the model search and reported fixed-model validation, the repeated scores may be mildly optimistic. The nested var1 result better accounts for tuning but still uses the available training dataset. A final untouched labeled holdout would be the strongest additional check.

10 Conclusion

Cross-validated selection produced two reproducible polynomial models. Degree-5 Lasso term selection followed by Ridge reduced var1's held-out error beyond Ridge alone. Degree-12 Ridge remained the best tested choice for var2. Both pipelines were retrained on all available labeled observations and used to create the final prediction files for BT2024128.